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a) Lets use the series from Q2c) for this.
Recall $X_t = Z_0 S_t + Z_1 S_{t-1}$.
We have already proven it is weakly stationary.

Since the seasonality has a period of 2, then we can conclude that $X_t = \begin{cases} Z_0 S_t & \text{when } t \text{ is even} \\ Z_1 S_{t-1} & \text{when } t \text{ is odd} \end{cases}$

As such, we need only assume that Z_0 and Z_1 have a different distribution for their random variables (e.g. $Z_0 \sim \mathcal{N}(0,1)$ and $Z_1 \in \{-1, 1\}$ w/ 0.5 prob. each).
If this is the case, then it is easy to see that X_t and X_{t+1} will not have the same joint distribution.
Even t will have $P(Z_0 = 1) = 0$ (property of normal distributions), and $P(Z_1 = 1) = 0.5$.
These two are different and immediately show that $\forall t$, $X_t \neq X_{t+k}$ for $k \in \mathbb{N}$, thus violating the definition of strong stationarity.

A1 Q3b

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b) No, if X_t is weakly stationary, X_t^2 is not necessarily weakly stationary.

As a counterexample, consider the time series from Q2b) once more, along with the definitions of $Z_0 \sim \mathcal{N}(0,1)$ and $Z_i \in \{-1, 1\}$ w/ prob. 0.5 each from Q3a).

For even t, when $X_t = Z_0 s_t$, note that $\text{Var}(X_t^2) = 2$ b/c $\text{Var}(Z_0^2) = 2\sigma^4$ (according to Math Stack Exchange Question 620045)

$= 2(1)$

$= 2$

For odd t, when $X_t = Z_i s_{t-1}$, $\text{Var}(X_t^2) = 0$ b/c the probability of this simple discrete distribution doesn't change.

Given that these variances differ based on when t is even or odd, we have shown that the Covariance of X_t^2 relies on t. This violates the definition of weak stationarity.

Therefore we have shown a counterexample showing that X_t may be weakly stationary, but X_t^2 is not weakly stationary.